Daniel Fisher

Senior Software Engineer, Data & AI

Contact

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daniel-fisher

Education

PhD Remote Sensing 2009 – 2014
University College London

MSc Remote Sensing 2008 – 2009
University College London

BSc Geography 2003 – 2006
University College London

Certifications

AWS Certified Cloud Practitioner 2023

S2DS London 2016 Sep 2016
Pivigo
ID: Accredible-10278261

Profile Summary

Architected high-performance data infrastructure and serverless pipelines. Generated >80% of committed code via AI-assisted workflows (e.g., Cursor, Gemini CLI). Processed Petabyte-scale datasets and optimised complex transformations using Python and Rust. Built agentic platforms for model risk management and implemented robust ETL pipelines. Translated domain-specific geospatial metrics into universally understood data engineering solutions.

Skills & Abilities

Languages
Python, Rust, SQL

Data & Cloud
AWS (Batch, Step Functions, Lambda, S3), Ray, Zarr v3, Parquet, Serverless Pipelines, High-Performance Pipelines

Geospatial
Remote Sensing, Image Processing, Geospatial Analytics, STAC, Overpass API, Datashader

AI/ML
PyTorch, LangChain, LangGraph, Agentic Platforms, Computer Vision, GANs, Boosting, Auto-ML

Domain Knowledge
Fire Radiative Power (FRP), Gas Flare Radiative Energy, Smoke Plume Dynamics, Satellite Sensors (Himawari-8, Meteosat, ATSR-1/2, AATSR, SLSTR, CALIPSO, MISR, SEVIRI)

Professional Experience

Climate X

Senior Data Engineer Jul 2024 – Present

- **Scalable Pipeline Development:** Architected the next-generation data provisioning engine, a highly parallel, serverless pipeline using AWS Batch Array Jobs to convert raw climate data into optimised Zarr v3 multi-scale arrays. Previously scaled the legacy Ray orchestration engine across massive distributed workloads.
- **High-Performance Processing:** Optimised massive data transformations using high-performance Python tooling backed by Rust binaries, implementing zero-copy inplace array mutation and streaming Zarr validation to process Petabyte-scale datasets and eliminate out-of-memory crashes.
- **ML Model Monitoring & Evaluation:** Built an agentic platform for Model Risk Management using LangChain/LangGraph to ingest evidence, run compliance checks, and automate gap reports for model behaviour.
- **AI Tooling & Internal Adoption:** Built internal CLI tools that automatically inject standardised system prompts, verification checklists, and autonomous skills into repositories to constrain LLM behaviour and enforce safety.
- **Data Integration for Safety:** Engineered complex ingestion connectors pulling unstructured context from across the organisation to ground LLM evaluations.
- **Self-Service Tooling:** Built a complete, self-service QC orchestration engine via Slackbot integration, enabling non-engineers to trigger massive, distributed QC runs directly from chat and generate high-resolution Datashader plots.
- **Data Quality & Telemetry:** Integrated real-time error tracking and telemetry across spatial pipelines, building intelligent out-of-core pipeline safety limits for resilience.

NatCen (National Centre for Social Research)

Associate (Freelance) Apr 2024 – Present

- **Classification Software:** Contributed classification software for NatCen's "Dividing Lines" voter classification, used in the BBC's "Undercover Voters" coverage.
- **National Enrichment Pipeline:** Built a national enrichment pipeline at the LSOA level combining OSM infrastructure features with STAC satellite data, NASA DEMs, and ESA WorldCover land cover.
- **Geospatial ETL:** Implemented robust ETL pipelines with Overpass API query generation, retries/backoff, endpoint failover, caching, and resumable partial outputs, delivering national outputs for 35,672 LSOAs.

SatVu

Geospatial Engineer Feb 2022 – Jul 2024

- **Serverless Image-Processing:** Designed and shipped composable serverless image-processing pipelines; automated deployments via CI/CD.
- **Event-Driven Workflows:** Deployed event-driven APIs/workflows using Amazon S3 triggers orchestrated by AWS Step Functions and executed with AWS Lambda.
- **Computer Vision & ML:** Applied ML/computer vision in PyTorch, including GAN-based cloud segmentation (improved IoU from 0.4 to 0.6) and D2-Net image registration (improving alignment from 100m+ to ~15m vs raw satellite metadata).

King's College London / NCEO

Research Associate Jun 2014 – Feb 2022

- **Large-Scale EO Pipelines:** Built large-scale Earth Observation processing pipelines (>100TB / ~800k images) for long-horizon monitoring applications.
- **LSA SAF:** Developed algorithms and product contents for Meteosat Fire Radiative Power (FRP) products used in Copernicus Atmosphere Monitoring Service (CAMS).
- **Global Gas Flaring Inventory:** Conducted multi-decade global inventories of gas flaring using ATSR and SLSTR satellite records.

Selected Publications

- *Multi-decade global gas flaring change inventoried using the ATSR-1, ATSR-2, AATSR and SLSTR data records* (2019).
- *Shortwave IR adaption of the mid-infrared radiance method of FRP retrieval for assessing industrial gas flaring output* (2018).
- *Major advances in geostationary fire radiative power (FRP) retrieval over Asia and Australia stemming from use of Himawari-8 AHI* (2017).
- *Automated stereo retrieval of smoke plume injection heights... from AATSR* (2014).

For a full list of publications, please visit my Google Scholar profile.